

Modern Application Development-2

Project Report

TrekGenie - Trekking Management Application

1. STUDENT DETAILS

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Project name	TrekGenie - Trekking Management System
About Me:	I am a software developer and entrepreneur currently pursuing the BS Degree in Data Science and Applications at IIT Madras. My interests include full-stack web development, data science, and AI-driven applications that solve practical problems.

2. Project Details

Project Title:

TrekGenie - Trekking Management Application

Problem Statement:

The objective of this project was to design and develop a modern full-stack Trek Management System that enables administrators to efficiently manage trekking expeditions, assign staff members, monitor bookings, and oversee registered users. The application also provides trekkers with a seamless platform to browse available trekking expeditions, make bookings, monitor payment status, receive automated reminders, and export their trekking history.

The project demonstrates concepts such as RESTful API development, role-based authentication and authorization, asynchronous task processing, caching, and responsive web application development.

Approach:

The application follows a modular architecture consisting of a Flask backend and a Vue.js frontend communicating through REST APIs. JWT-based authentication secures all protected resources while Role-Based Access Control (RBAC) restricts access to features based on user roles (Administrator, Staff, and Trekker).

Redis caching has been integrated to improve application performance by caching frequently accessed trek information. Celery and Celery Beat have been used to execute asynchronous background jobs including daily reminder emails, monthly activity reports, and user-triggered CSV exports.

The frontend was developed using Vue.js 3 with Vue Router, Axios, and Bootstrap 5 to provide a clean and responsive user interface. SQLAlchemy ORM manages database interactions while SQLite serves as the relational database.

3. AI/LLM Declaration

Artificial Intelligence tools (ChatGPT – GPT-5) were used throughout the project primarily for brainstorming software design, reviewing implementation strategies, debugging runtime issues, improving user interface design, generating boilerplate code, and assisting with documentation.

The approximate extent of AI-assisted development is estimated to be **25–30%**. All architectural decisions, application logic, database design, debugging, testing, feature integration, and final implementation were carried out manually by the author. Every AI-generated suggestion was reviewed, modified where necessary, and fully understood before being incorporated into the project.

4. Technologies and Frameworks Used

Technology	Purpose
Flask	Backend REST API Framework
SQLAlchemy	Object Relational Mapper (ORM)
SQLite	Relational Database
Flask-JWT-Extended	JWT Authentication
Flask-Caching	Redis Cache Integration
Redis	API Response Caching
Celery	Asynchronous Background Tasks
Celery Beat	Scheduled Background Jobs
Flask-CORS	Cross-Origin Resource Sharing
Vue.js 3	Frontend Framework
Vue Router	Client-side Routing
Axios	HTTP Client
Bootstrap 5	Responsive User Interface
MailHog	Email Testing Server

5. Database Schema / ER Diagram

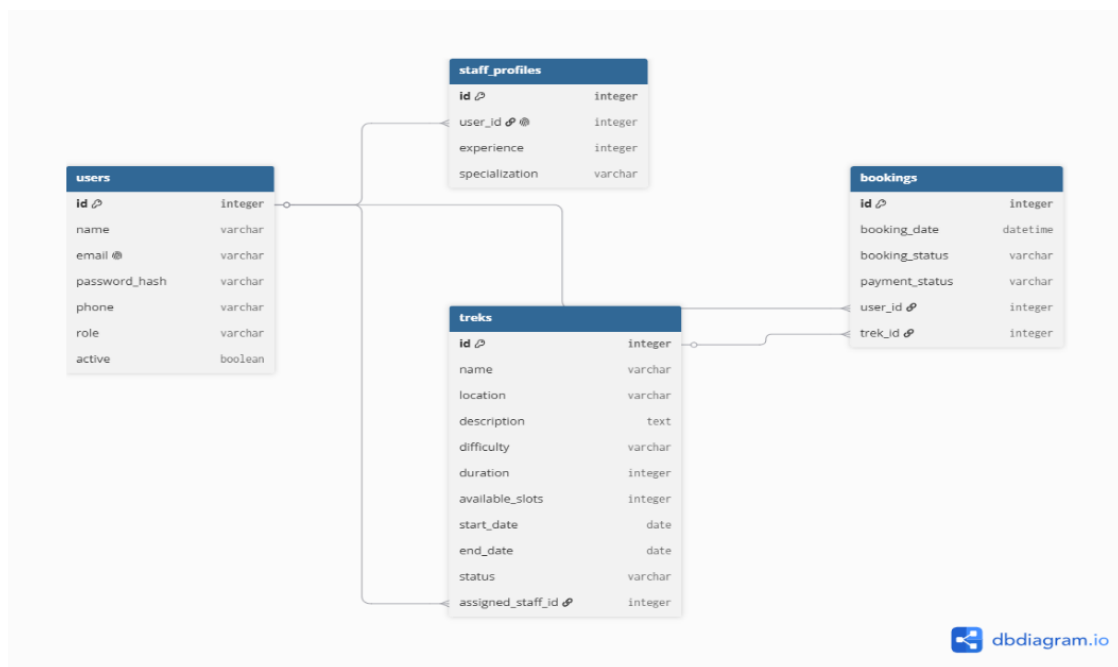
The application consists of four primary entities.

User	Trek	Fields	StaffProfile
Stores all registered users including Administrators, Staff Members, and Trekkers.	Stores complete information about trekking expeditions.	Stores all trek bookings made by trekkers.	Stores additional information about staff members.
Fields • id • name • email • password_hash • phone • role	Fields • id • name • location • description • difficulty • duration • available_slots	Fields • id • booking_date • booking_status • payment_statuses • user_id	Fields • id • user_id • experience • specialization

• active	• start_date • end_date • status • assigned_staff_id	• trek_id	
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Relationships

- One Staff Member can manage multiple Treks.
- One Trek is assigned to one Staff Member.
- One Trekker can create multiple Bookings.
- One Trek can have multiple Bookings.
- One Staff Member has exactly one Staff Profile.



6. API Resource Endpoints

Endpoint	Method	Description
/auth/register	POST	Register a new trekker
/auth/login	POST	Authenticate user
/admin/dashboard	GET	Admin dashboard statistics
/admin/staff	GET / POST	Manage staff
/admin/users	GET	View registered users
/admin/treks	GET / POST	Manage trekking expeditions
/admin/bookings	GET	View all bookings
/staff/dashboard	GET	Staff dashboard
/staff/treks	GET	Assigned treks
/trekker/dashboard	GET	Trekker dashboard
/trekker/treks	GET	Browse available treks
/trekker/book	POST	Book a trek
/trekker/bookings	GET	View booking history
/trekker/export	POST	Export booking history

8. Video Presentation

Drive link	Click here for the demo video
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